MIL-V-44139(GL)
27 March 1985

MILITARY SPECIFICATION

VERMICELLI WITH MEAT AND SAUCE, COOKED, DEHYDRATED - (OPERATIONAL RATION COMPONENT)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers dehydrated, cooked vermicelli with meat and sauce for use by the Department of Defense as a component of the Food Packet, Assault.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-44146 - Packaging of Food Packet Assault Components in Flexible Bags

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

U.S DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standard of Identity for Parmesan Cheese

(Application for copies should be addressed to Director, Dairy Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Tomato Puree
U.S. Standards for Grades of Tomato Paste

(Application for copies should be addressed to Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Washington, DC 20250.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

THE UNITED STATES PHARMACOPOEIAL CONVENTION (USP)

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. Beef shall be the only meat ingredient used and shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters and any of the following recognizable primal and subprimal cuts: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, knuckles, and bottom rounds. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used except tenderloins and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable; and there shall be no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The fresh chilled beef shall be boned and trimmed to remove and exclude objectionable material such as a bone, cartilage, heavy connective tissue, etc. The boneless trimmed meat shall meet the limitations specified in table I and table II.

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3.2.1.2 Handling and storage. Handling and storage of boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the same day as boning and trimming shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for up to 4 days is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Beef shall not be frozen at any time prior to further processing of the boned and trimmed beef.

3.2.2 Allspice, ground. Ground allspice shall be derived from the dried, nearly ripe fruit of Pimenta officinalis L. and shall have a fragrant, clove-like aroma, strongly aromatic, pungent, clovelike flavor and a dark reddish brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 25 sieve.

3.2.3 Basil, ground. The ground basil shall be from the dried leaves of Ocimum basilicum L. and shall possess a sweet anise-like odor and aromatic, warm, slightly pungent flavor. Basil shall contain not less than 0.4 mL of volatile oil per 100 grams and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 35 sieve.

3.2.4 Bay leaves, ground. Ground bay leaves shall be from the dried leaves of Lauris nobilis L. and shall have a delicate aromatic odor and aromatic bitter flavor. Bay leaves shall contain not less than 1.0 mL of volatile oil per 100 grams and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.5 Cheese, parmesan, grated. The parmesan cheese shall be grated and conform to the Standards of Identity for Parmesan Cheese. The cheese shall be aged 10 months or more and shall be free from musty, oxidized, tallowy or unclean flavors. It shall be free flowing and free from lumps except those that break up readily with slight finger pressure.

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3.2.6 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.7 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.8 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor and a light color. The ground white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.9 Salt. Salt shall be non-iodized, white, refined sodium chloride, with or without anti-caking agents.

3.2.10 Oregano, ground. Ground oregano shall be derived from the dried leaves of Origanum vulgare L. and shall have a strong camphoraceous aroma and a pungent slightly bitter flavor. The ground oregano shall contain not less than 2.0 mL of volatile oil per 100 grams and shall be of such size that 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.11 Vermicelli. Vermicelli shall be unenriched, 0.040 to 0.059 inch solid cylindrical type. The vermicelli shall be made from only semolina and shall contain distilled glycerolmonostearate. The vermicelli shall be 3 to 6 inches in length.

3.2.12 Tomatoes, canned, crushed, concentrated. Canned crushed tomatoes shall be high quality peeled crushed tomatoes packed in puree with the solids content not less than 10.0 percent.

3.2.13 Tomatoes, canned, pureed (tomato pulp). The pureed tomatoes shall be medium to heavy concentration (10.2 to 15.0 percent solids), coarse textured, U.S. Grade A of the U.S. Standards for Grades of Tomato Puree.

3.2.14 Tomato paste. The tomato paste shall be U.S. Grade A, light concentration (24 to 28 percent solids) of the U.S. Standards for Grades of Tomato Paste.

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3.2.15 Water. Water used for formulation, ice making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.16 Nitrogen. Nitrogen shall be U.S. Pharmacopoeia grade and shall be water or liquid nitrogen pumped.

3.3 Preparation and processing.

3.3.1 Beef preparation. The beef shall be prepared as follows:

3.3.1.1 Grinding of beef. The boned and trimmed meat shall be ground once through a plate having holes 1 to 1-1/2 inch in diameter (coarse grinding). The beef shall then be ground through a plate with holes 3/8 inch in diameter. The beef shall not exceed 50°F upon completion of either coarse or final grinding. The beef shall meet the time-temperature limitations specified in 3.3.1.4. Raw beef once frozen shall not be refrozen.

3.3.1.2 Fat content. The fat content of the ground beef shall not be less than 12 percent or more than 18 percent.

3.3.1.3 Frozen storage of ground beef. Frozen ground beef shall be packaged and packed to protect it from freezer deterioration or other damage. Containers used for packing shall be of the kind used commercially, provided they are adequate to maintain product excellence during storage.

3.3.1.4 Time-temperature limitations. The raw ground beef shall be handled so as to comply with the following limitations:

a. Maximum holding temperature and time of coarse ground beef:

1. If coarse ground beef is to be frozen	0°F 1/	30 days
2. If coarse ground beef is processed on the same day	50°F	24 hours 2/
3. If coarse ground beef is held chilled	40°F	72 hours 3/

b. Maximum holding temperature and time of final ground beef:

1. If final ground beef is from coarse ground beef that was previously frozen:	40°F	24 hours 4/
2. If final ground beef is from coarse ground beef processed on the same day	50°F	24 hours 2/

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3. If final ground beef is from chilled held coarse ground beef	40°F	72 hours 3/
4. If final ground beef is to be frozen	0°F 1/	30 days

1/ The ground beef shall be frozen to 0°F within 72 hours after placement in the freezer.

2/ Both the coarse and final grinding processes shall be done within 24 hours after boning and trimming.

3/ The beef may be held at 40°F provided the coarse and final grinding processes are completed within 72 hours after boning and trimming.

4/ Previously frozen coarse ground beef may be used provided the coarse ground beef is in the final grinding process with 24 hours after removal from the freezer.

3.3.2 Preparation of the vermicelli with meat and sauce. The vermicelli with meat and sauce shall be prepared as preconcentrated and formed slices or as compressed bars. Slice preparation shall be as outlined by 3.3.2.1, 3.3.2.2 and 3.3.2.3. Bar preparation shall be as outlined by 3.3.2.1, 3.3.2.2 and 3.3.2.4.

3.3.2.1 Vermicelli with meat and sauce formulation. The vermicelli with meat and sauce shall be formulated as follows:

Ingredient	Percent by weight
Beef, raw, ground	45.0
Tomatoes, crushed or pureed (10 to 15 percent solids)	23.0
Vermicelli, dry (3-6 inch cut)	15.0
Tomato paste (24 to 28 percent solids)	2.0
Seasoning mix	5.0
Water 1/	10.0

Seasoning mix

Cheese, parmesan, grated	45.0
Salt	22.0
Onion powder	15.0
Garlic powder	14.0
Oregano, ground	1.2

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Seasoning mix (cont'd)

Pepper, white, ground	1.0
Allspice, ground	1.0
Basil, ground	0.6
Bay leaves, ground	0.2

1/ For preconcentrated and formed slices, the formula water shall be reduced by an amount that will yield a product complying with the finished product requirements.

3.3.2.2 Mixing and cooking. The vermicelli, ground beef, and other ingredients shall be mixed and cooked as follows:

a. Cook vermicelli in boiling water until just cooked (firm, with visible dense center). Drain quickly and rinse with cold water to stop cooking and remove starch.

b. The water, ground beef, and seasoning mix shall be mixed and heated until the redness of the meat disappears.

c. The tomato paste and crushed or pureed tomatoes shall be added to the meat and the product heated to 160°F.

d. Add the cooked vermicelli and stir gently to avoid further breaking up of the vermicelli.

3.3.2.3 Slice preparation. The preconcentrated and formed slices shall be prepared as follows; or alternatively, the cooked product may be formed so that the dimensions, weights, and volumes specified in 3.4(e) are met. The times and temperatures specified below shall be followed according to the process being used.

a. The cooked product shall be immediately mechanically stuffed into suitable casings. Casings shall have an approximate 5-1/2 inch lay-flat dimension.

b. Each stuffed casing shall be formed (see 6.3) into a rectangular shape in accordance with 3.4(e).

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c. The formed product shall be placed in a freezer within 4 hours after stuffing and frozen to an internal temperature of 0°F or below within 24 hours after placing in the freezer. Frozen product shall not be held for longer than 30 days before slicing and freeze dehydration.

d. The frozen product shall be cut into slices so that four dehydrated slices will meet the requirements specified in 3.4(e).

e. The slices shall be placed in the freezer dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 24 hours after slicing. The product shall be protected at all times from contamination and freezer burn.

3.3.2.4 Bar preparation. The compressed bars shall be prepared as follows:

a. The cooked product shall be portioned onto freeze dryer trays.

b. The product shall be placed in a freezer within 4 hours after cooking and frozen to 0°F or lower within 12 hours after placing in the freezer. The product shall be protected at all times from contamination and freezer burn.

c. The product shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 48 hours after placing in the freezer.

d. The dehydrated product shall be conditioned by adding 10 percent by weight of water. The moistened product shall be processed in a microwave unit at the equivalent of 30 seconds at 1000 watts (see 6.4).

e. The conditioned product shall be compressed into bars 1 inch wide by 3 inches long by not more than 1-1/4 inches thick so that three dehydrated bars will meet the requirements specified in 3.4(f). The compressed bars shall have edges and corners rounded to a nominal 1/8 inch curvature. The product shall be compressed within 5 days of initial freeze dehdyration.

f. The compressed bars shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehdyration shall be started within 48 hours of compression.

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3.3.3 Dehydration. The vermicelli with meat and sauce shall be freeze dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury and at a product temperature as indicated by suitable instruments, not to exceed 150°F. If the platen temperature is maintained at 155°F or below with radiant heating, the product temperature may be disregarded. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen. In no case shall more than 16 hours elapse between dehydration and completion of packaging. During this interim period, product shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches of mercury for the entire period. If vacuum is used, it shall be broken with nitrogen. For compressed bar product during the interim between freeze drying and compression the product shall also be protected as above.

3.3.4 Pouch filling and sealing. Each pouch shall be filled and sealed such as to conform to the following requirements:

a. Three bars or four slices shall be inserted into a mixing bag conforming to the requirements of MIL-P-44146.

b. The bag closure device (see MIL-P-44146) shall be placed within the outerfolds of the mixing bag so as to minimize any protrusion against the pouch.

c. The filled mixing bag shall be folded and fitted into the pouch.

d. The pouch shall be sealed in accordance with the requirements of MIL-P-44146.

3.4 Finished product. The finished product shall comply with the following requirements.:

Dehydrated product

a. There shall be no foreign material, such as but not limited to dirt, glass, paint, hard plastic, wood, insects, insect parts or noxious chemicals. There shall be no innocuous soft extraneous material (paper, string, etc.), smears or stains (oil, grease, etc).

b. There shall be no foreign odor or flavor such as, but not limited to burnt, scorched, stale, sour, rancid, musty, or moldy.

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c. There shall be no color foreign to the product.

d. There shall be no evidence of incomplete dehydration.

e. When the product is in the form of slices:

1. Each pouch shall contain four slices.
2. Each slice shall be approximately 2.25 ± 0.1 by 3.25 ± 0.1 by 1/2 to 3/4 inches.
3. The volume of the four slices shall be not greater than 246 cubic centimeters.
4. The total weight of the four slices shall be not less than 90 grams and not greater than 100 grams.

f. When the product is in the form of bars:

1. Each pouch shall contain three bars.
2. Each bar shall be 3 ± 0.1 by 1 ± 0.1 by not greater than 1-1/4 inches.
3. The total weight of the three bars shall be not less than 90 grams and not greater than 100 grams.
4. Each bar shall have rounded edges and corners.

g. The product shall be sufficiently compressed or formed so that it can be packaged without breaking or crumbling.

h. Moisture content of the dehydrated product shall not exceed 2.0 percent.

i. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

j. The E coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample.

Rehydrated product

k. When the product is rehydrated as specified in footnote 3 of table III, the product shall show not less than 80 percent rehydration.

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l. The rehydrated vermicelli with meat and sauce shall possess flavor, odor and texture considered normal for frozen and heated vermicelli with meat and sauce. There shall be no scorched, burnt, or foreign flavors or odors.

m. When rehydrated, the vermicelli with meat and sauce shall show distinct strands of vermicelli and pieces of meat. The product shall not be a mass of indistinguishable materials.

3.4.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The meat component and finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agriculture Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

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TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories Major AC	Major RE	Minor AC	Minor RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/
Major	Minor	
151		Presence of lymph gland measuring 1.0 inch or more in any dimension.
	201	Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
152		Presence of blood clot measuring 1.0 inch or more in any dimension.
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
153		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
154		Presence of bone measuring 0.3 inch or more in any dimension.
155		Presence of cartilage measuring 0.5 inch or more in any dimension.
156		Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
157		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.

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TABLE II Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3 square inches or more.
158		Presence of knuckle cover.
159		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, prepubic tendon, hanging tender, or thymus gland measuring less than 1.0 inch in longest dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	213	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	214	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney, or pelvic fat.

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be determined by a Meat and Poultry Inspection Program employee.

2/ Evidence of freezing or defrosting or presence of any off-condition, (including but not restricted to being off-odor, slightly sticky, gassy, rancid, or sour, or showing discoloration) shall cause rejection of the involved lot.

4.5.1.3 Other ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, handling and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

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4.5.2.1 In-process testing for fat content. Four 1-pound samples shall be selected at random from the day's production of raw ground beef. Each of the four samples shall be tested for fat content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. Results shall be reported to the nearest 1 percent. If the fat content of one or more samples is greater than the limit specified in 3.3.1.2, the lot of ground beef or any product made therefrom shall be rejected.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44146.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch, mixing bag, and closure device. Any individual net weight of less than 90 grams or greater than 100 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 1 gram.

4.5.5 Volume test. (slices only). Test for volume of slices shall be performed as follows:

Equipment

a. Container, large enough to submerge pouches in; having a spout approximately 1 inch below the container surface.

b. 150 gm weight with a clip attached.

c. 250 mL or 500 mL beaker.

d. 250 mL or 500 mL graduated cylinder.

e. Vacuum-sealed pouch containing a mixing bag and a closure device.

Procedure

a. Establish tare as follows:

(1) Place container on a level, vibration-free surface.

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(2) Fill the container with water until the water comes out the spout. Wait until the water stops dripping before proceeding.

(3) Position the beaker under the spout.

(4) Attach the vacuum-sealed pouch containing a mixing bag and a closure device to the 150 gm weight with the clip. Submerge the pouch with the attached weight into the filled container.

(5) Allow the water to stop dripping before measuring, approximately 5 minutes.

(6) Measure the collected water in the graduated cylinder and record this measurement as the volume of the tared pouch. Report measurement to the nearest 2 mL.

(7) Empty and dry the beaker and graduated cylinder.

b. Testing of filled and sealed pouch:

(1) Refill the container and, using the same procedure as above, test the filled and sealed pouch.

(2) Measure the collected water in the graduated cylinder and record this measurement as the volume of the filled and sealed pouch. Report measurement to the nearest 2 mL.

c. Determine the volume of the slices using the following equation:

Volume of filled and sealed pouch	minus	Volume of the tared pouch	=	Volume of the slices

The lot size shall be expressesd in filled and sealed pouches. Any result not conforming to the requirement in 3.4(e) shall be scored as a major defect. The sample size shall be the number of pouches indicated by inspection level S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table III. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

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TABLE III. Product defects 1/ 2/

Category		Defect
Major	Minor	
151		Mixing bag missing.
152		Number of bars/slices in pouch not as specified.
	201	Bag closure device missing.
		Applicable to dehydrated product
153		Evidence of incomplete dehydration.
	202	Dimensions of compressed bars not as specified.
	203	Broken or crumbled (more than 3 large pieces) slice or bar in a pouch.
	204	Edges and corners of compressed bars not rounded.
		Applicable to rehydrated product 3/
154		Product shows less than 80 percent rehydration. 4/
155		Product does not have distinct strands of vermicelli and pieces of meat (i.e. the product is a mass of indistinguishable materials).

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, noxious chemicals, rancid, sour, stale, etc.), foreign color or innocuous soft extraneous material, smears or stains (oil, grease, etc.) shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To perform rehydration examination, the entire contents of the pouch shall be used. Break bars or slices each into five or six pieces of comparable size. Add 180° to 200°F water to the broken product in the amount of 2.8 mL of water for each gram of dehydrated product and weigh. Stir to expose all surfaces to the water. Rehydrate for 5 minutes, stir, and rehydrate for an additional 5 minutes. Immediately after the 10 minute rehydration period, pour contents onto a U.S. Standard No. 40 sieve. Drain for 5 minutes and weigh.

4/ Percent rehydration shall be calculated as follows:

$$\frac{\text{Weight of rehydrated product}}{\text{Weight of dry product plus added water}} \times 100 = \text{Percent rehydration}$$

4.5.7 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Moisture-Drying in Vacuum method in the Meat section of the Meat and Meat Products chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists, except that a temperature of 70°C for 16 hours under a pressure of 100 mm of mercury shall be used. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4(h) shall be considered a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.8 Microbiological testing. Five filled and sealed pouches of finished product shall be selected at random from the lot regardless of lot size. The contents of each sample pouch shall be tested for aerobic plate count and E. coli in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Microbiological Methods: Section: Examination of Frozen, Chilled, Precooked, or Prepared Foods - Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of that sample. Continue as directed by AOAC. Any result not conforming to the requirements in 3.4(i) and (j) shall be considered a test failure. Any test failure shall be cause for rejection of the lot.

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4.5.9 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44146.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44146.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.4.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Forming of slices. The best results for forming the preconcentrated slices has been shown to be by placing the casings into the screen molds prior to stuffing. Molds produced by Beacon Metal Products, Inc., of Chicago, IL have been known to meet the requirements of this document.

6.4 Compression general. The best results for microwave heating have been achieved by placing the food in a closed container, e.g., a plastic bag, and placing the bagged food flat in the bottom of the oven. If the heating is done in 10 to 15 second cycles, turning the container, and mixing the food between cycles, more thorough distribution of the moisture will be achieved. Products with high fat content may require 1 to 2 minutes to cool before compression.

Custodian:

Army - GL

Preparing activity:

Army - GL

Review activities:

Army - MD, TS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-44139(GL)	Vermicelli with Meat and Sauce, Cooked, Dehydrated - (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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